ECE275 Midterm 1 Spring 2026

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Student Name: Student Email:

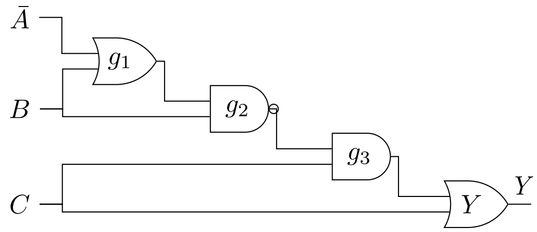
1 Instructions

- Time allowed is 75 minutes.
- In order to minimize distraction to your fellow students, you may not leave during the last 10 minutes of the examination.
- The examination is closed-book. One 8×11 in two-sided cheatsheet is allowed.
- Non-programmable calculators are permitted.
- The maximum number of marks is 75, as indicated; the midterm examination amounts 10% toward the final grade.
- Please use a pen or heavy pencil to ensure legibility. Colored pens/pencils are recommended for K-map grouping.
- Please show your work; where appropriate, marks will be awarded for proper and well-reasoned explanations.
- SOP stands for Sum of Products and POS stands for Product of Sums.

Problem 1. *Use the following 4-variable K-map for F (A, B, C, D), and find a minimal SOP expression for F (10 marks)*

CD \ AB	00	01	11	10
00	1	0	0	1
01	1	d	1	1
11	0	0	0	1
10	d	0	0	d

Problem 2. *Use Boolean algebra to find a simplified SOP expression for Y (10 marks)*



Problem 3. *Draw the ANSI diagram for the following function, using only NAND gates.* $F = \bar{A}B\bar{C} + BD + AC + \bar{B}C\bar{D}$. *(10 marks)*

Problem 4. *Consider a circuit to subtract two two-bit unsigned numbers. Denote the two bits of first number as A_1A_0 (forming the number N_A) and the two bits of second number as B_1B_0 (forming the number N_B). The circuit will find the difference $N_A - N_B$ which is denoted using three-bit two's complement notation $D_2D_1D_0$.*

(It is a bit unusual to represent differences of unsigned numbers in two's complement notation. It is more common to use the same notation type both on the inputs and on the outputs.)

About two's complement: *A 3-bit two's complement notation can represent numbers from -4 to $+3$. For positive numbers the notation stays the same ($0_{10} = 000_2$, $1_{10} = 001_2$, $2_{10} = 010_2$, $3_{10} = 011_2$). For negative numbers, the two's complement is obtained by subtracting the magnitude of the number from 8 (or equivalently by bitwise negation followed by addition of 1), leading to ($-1_{10} = 111_2$, $-2_{10} = 110_2$, $-3_{10} = 101_2$, $-4_{10} = 100_2$).*

1. *Start with filling out the following truth table (5 example rows are provided) (10 marks).*
2. *Use K-maps to find minimal Product of sum form for D_0 (15 marks).*
3. *Draw an ANSI network for D_0 using NOR gates only (5 marks).*

A_1	A_0	B_1	B_0	D_2	D_1	D_0
0	0	0	0	0	0	0
0	0	0	1	1	1	1
0	0	1	0	1	1	0
0	0	1	1	1	0	1
0	1	0	0			
0	1	0	1			
0	1	1	0			
0	1	1	1			
1	0	0	0			
1	0	0	1			
1	0	1	0			
1	0	1	1			
1	1	0	0			
1	1	0	1			
1	1	1	0			
1	1	1	1	0	0	0

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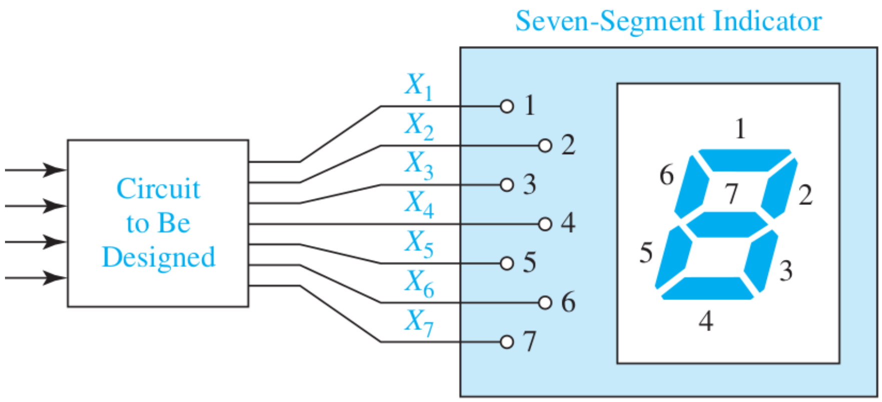
Problem 5. *You are provided a seven-segment display which takes seven inputs $X_1, X_2, \ldots, X_7$ for the segments as labeled below:*



Assume that the segment lights up when the signal is 0 *(low/off).*

Design a SOP circuit for X_1 which displays the letters A through J on a seven-segment indicator. The circuit has four inputs W, X, Y, Z which represent the last 4 bits of the ASCII code for the letter to be displayed. The ASCII codes for the letters are such that, if WXYZ = 0001, "A" will be displayed, for WXYZ = 0002, "b" will be displayed, and so on for WXYZ = 1010, "J" will be displayed. The letters should be displayed on the seven segment display in the following form (15 marks):

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